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```
def _detrnd(input_signal, lambda_value):
    """Detrnd PPG signal."""
    signal_length = input_signal.shape[0]
    # observation matrix
    H = np.identity(signal_length)
    ones = np.ones(signal_length)
    minus_twos = -2 * np.ones(signal_length)
    diags_data = np.array([ones, minus_twos, ones])
    diags_index = np.array([0, 1, 2])
    D = spdiags(diags_data, diags_index,
                (signal_length - 2), signal_length).toarray()
    detrnded_signal = np.dot(
        (H - np.linalg.inv(H + (lambda_value ** 2) * np.dot(D.T, D))), input_signal)
    return detrnded_signal
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```
├── physnet_optimizations.py
├── benchmark/
│   ├── run_benchmark.py
├── results/
│   ├── benchmark_results.csv
│   └── environment.json
├── report/
│   ├── report.md
│   ├── report.docx
│   └── report.pdf
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